

Quantum topology and VOA structures

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Vertex algebras, automorphic forms and combinatorics
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S. Gukov



P. Orland



D. Passaro



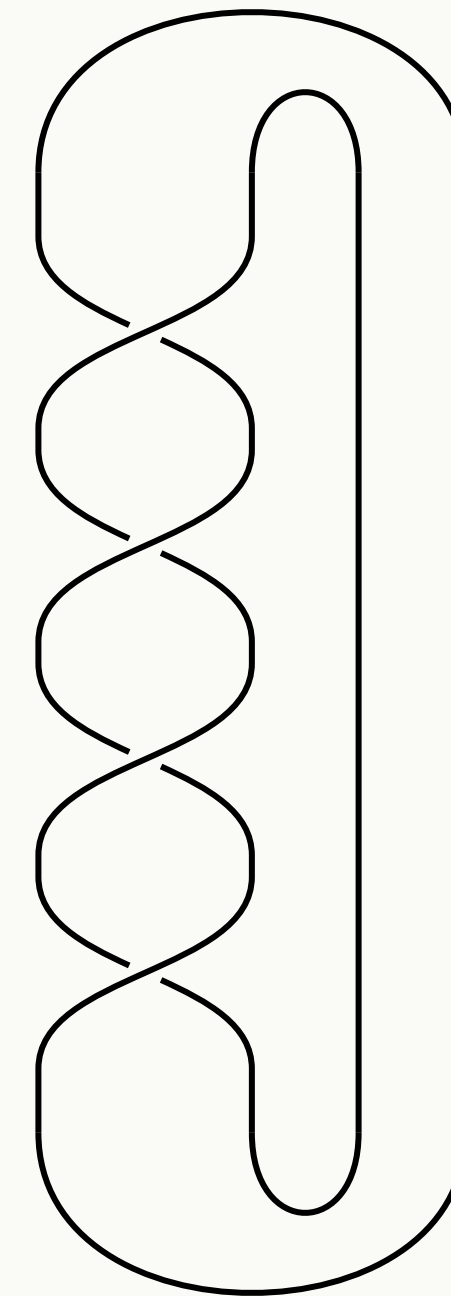
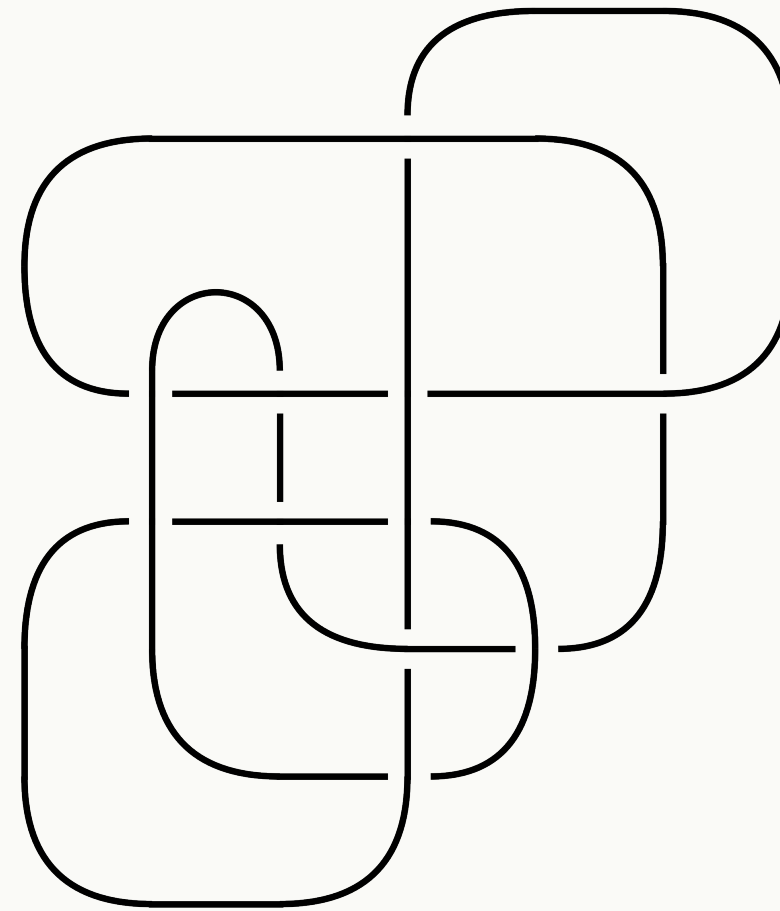
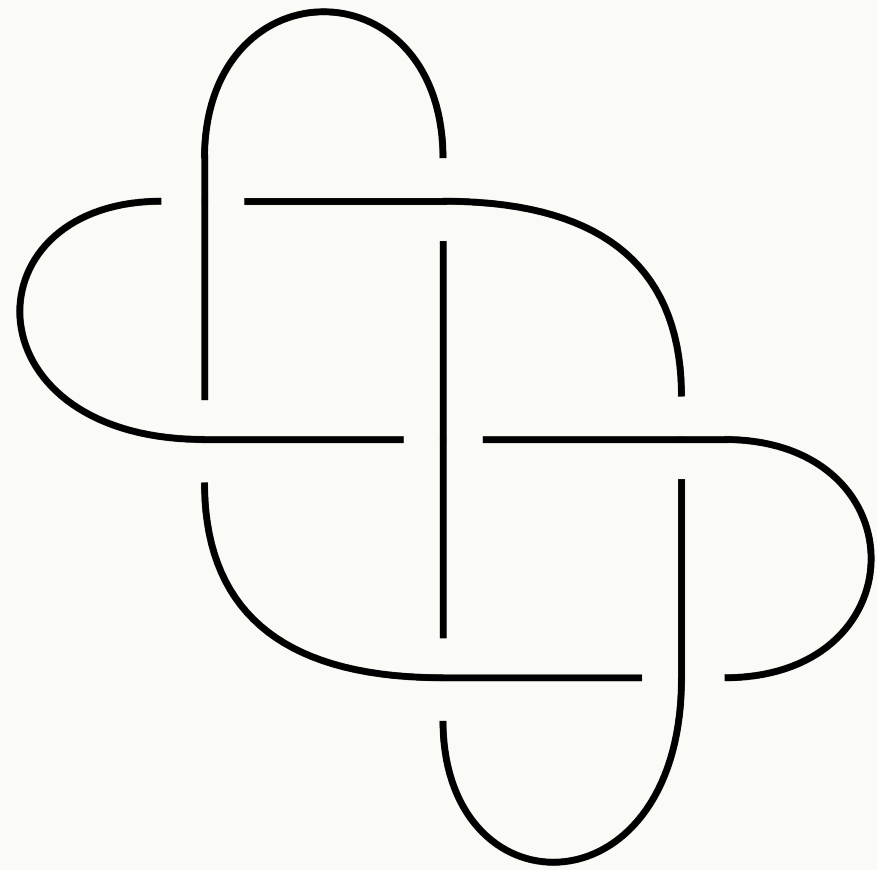
T. Saunders-A'Court



J. Svoboda

Knots and 3-manifolds

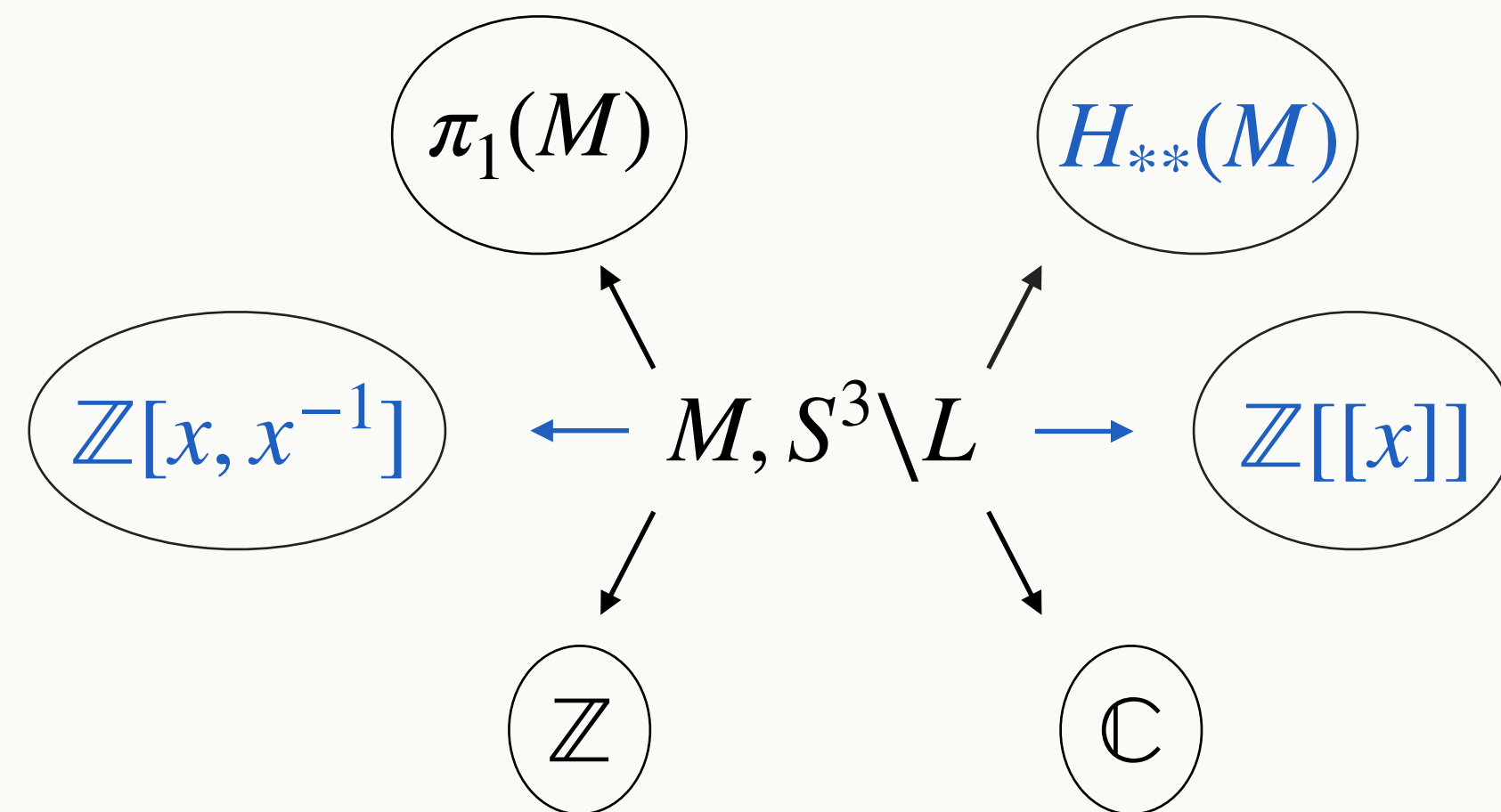
Knots are embeddings $S^1 \hookrightarrow S^3$. Links are embeddings $S^1 \cup S^1 \cup \dots \cup S^1 \hookrightarrow S^3$.



Knots and 3-manifolds

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Theorem (Lickorish—Wallace): Every closed, orientable 3-manifold can be obtained from a link by doing integral Dehn surgery.



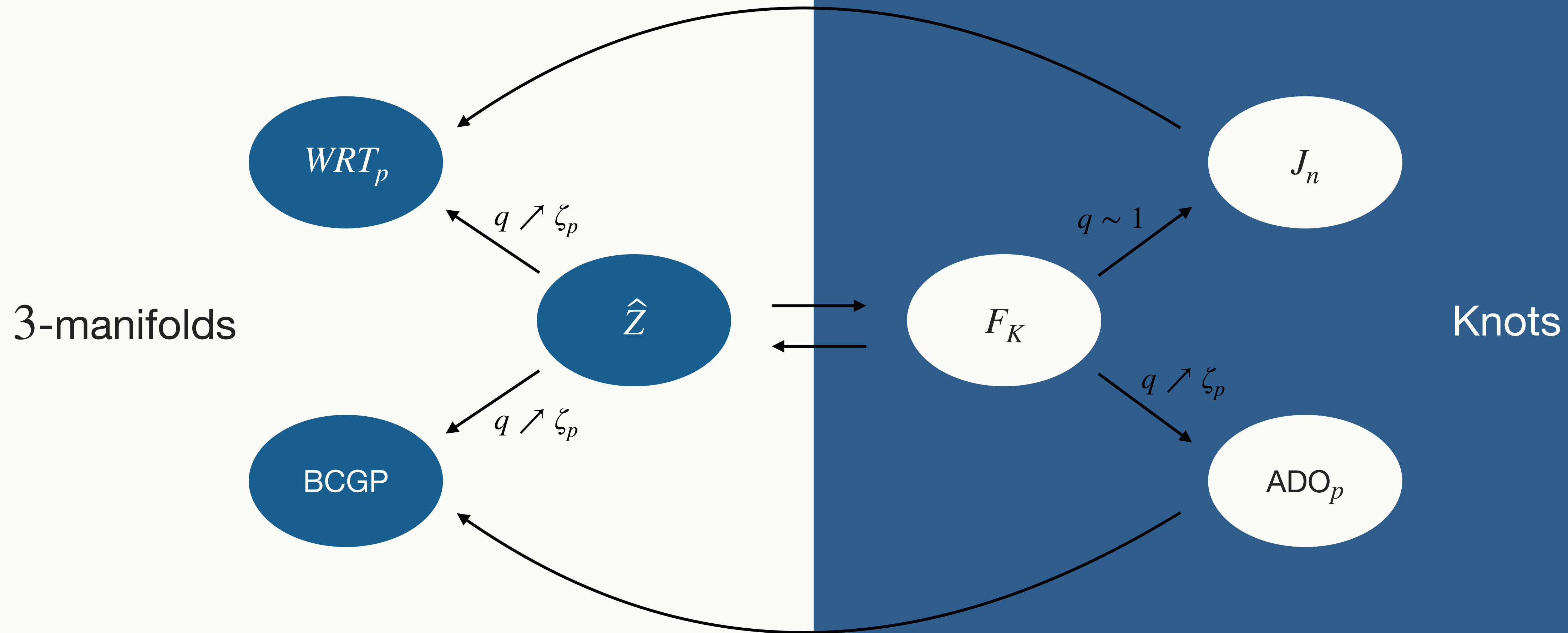
Invariants derived from the representation theory of quantum groups $\rightsquigarrow$ quantum topology.

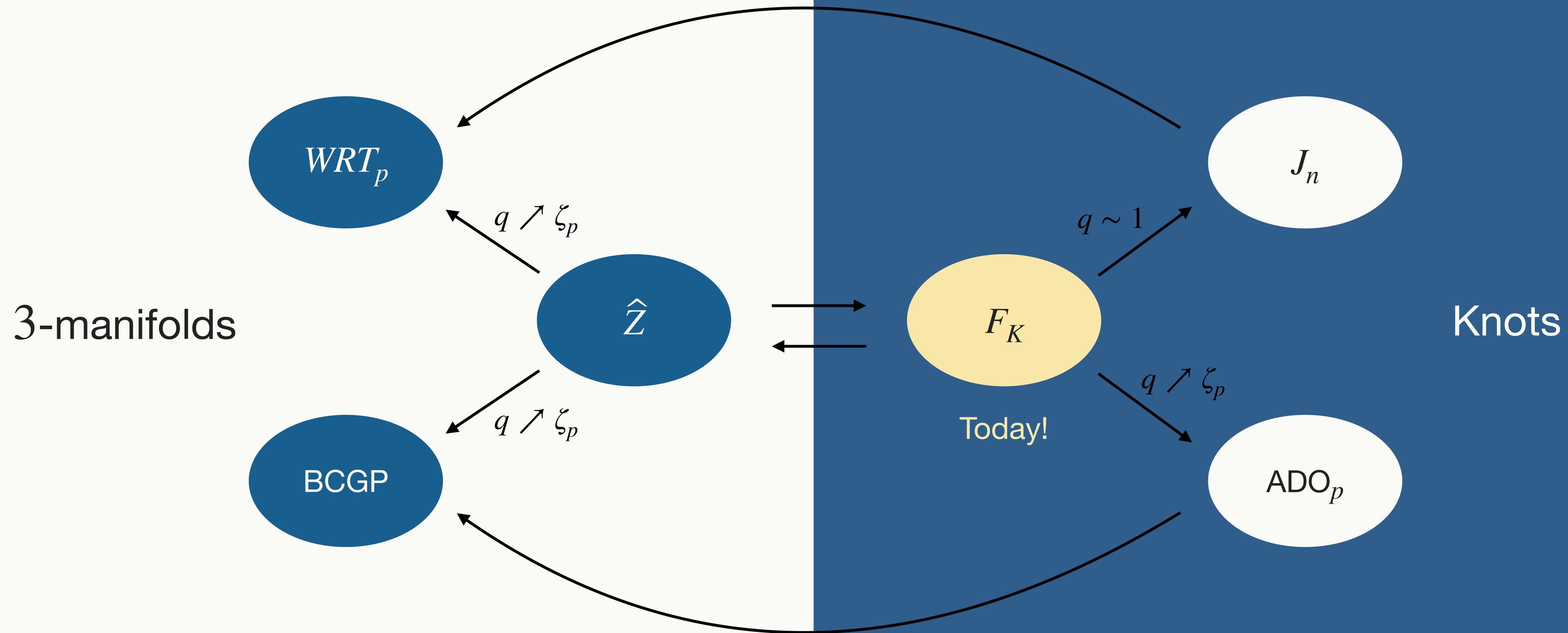
Invariant	Quantum group construction	VOAs
WRT	Quantum group: $U_q(\mathfrak{sl}_2)$ at $q = e^{2\pi i/p}$ Category: $U_q(\mathfrak{sl}_2)\text{--Rep}^{\text{S.S.}}$	$SU(2)_p$ WZW model

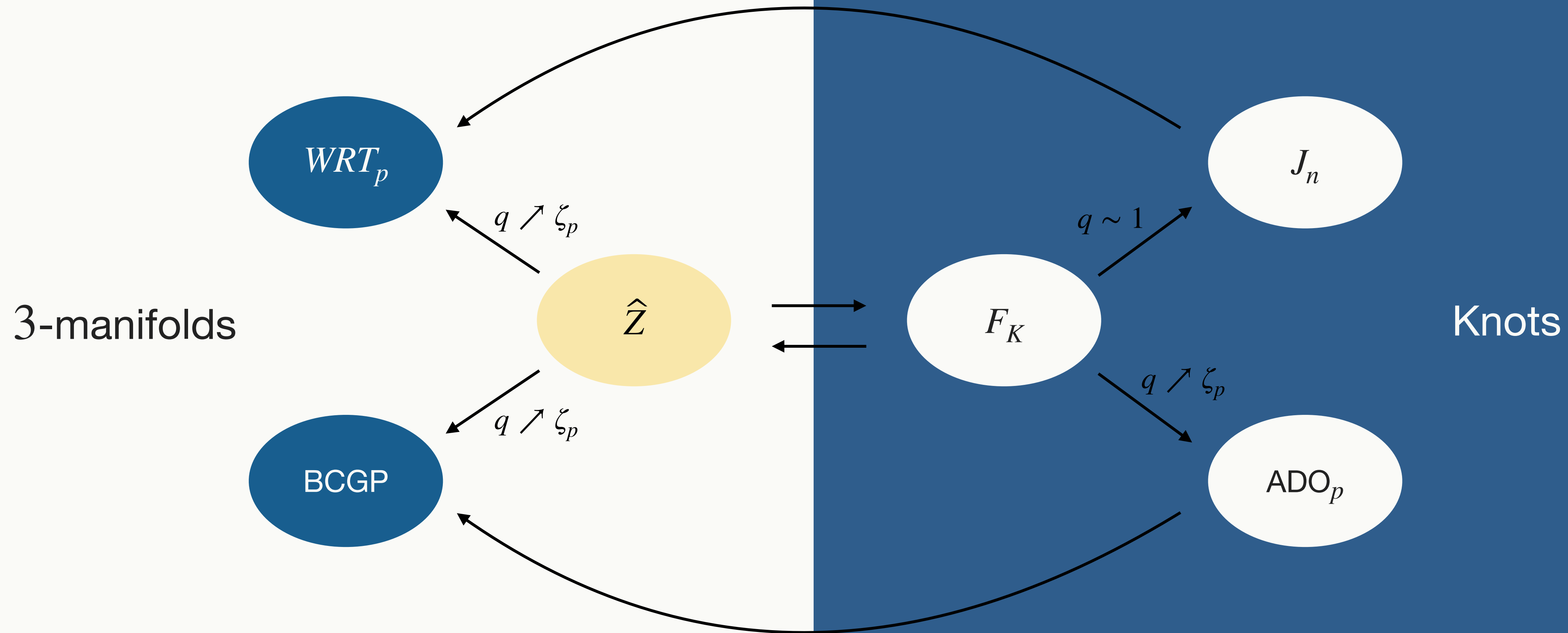
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GPPV $\hat{Z}(q) \in q^\Delta \mathbb{Z}[[q]]$ GM $F_K(x, q) \in \mathbb{Z}((q))[[x]]$	Quantum group: $U_q(\mathfrak{sl}_2)$, generic q Category: ?	?

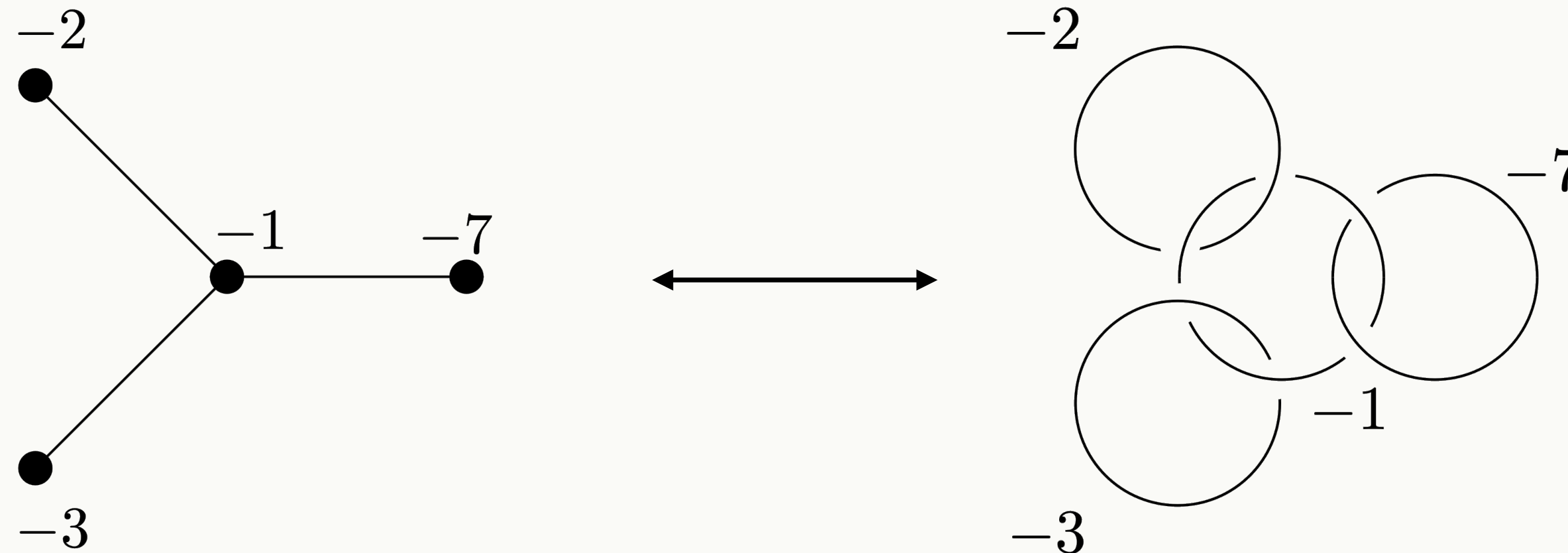






$\hat{Z}$ –invariant

- Invariant of 3-manifolds defined by **Gukov—Pei—Putrov—Vafa**.
- Defined for a class of “combinatorial” 3-manifolds, called **plumbed manifolds**.



Weighted graph (Γ_M, Q)

Surgery manifold M

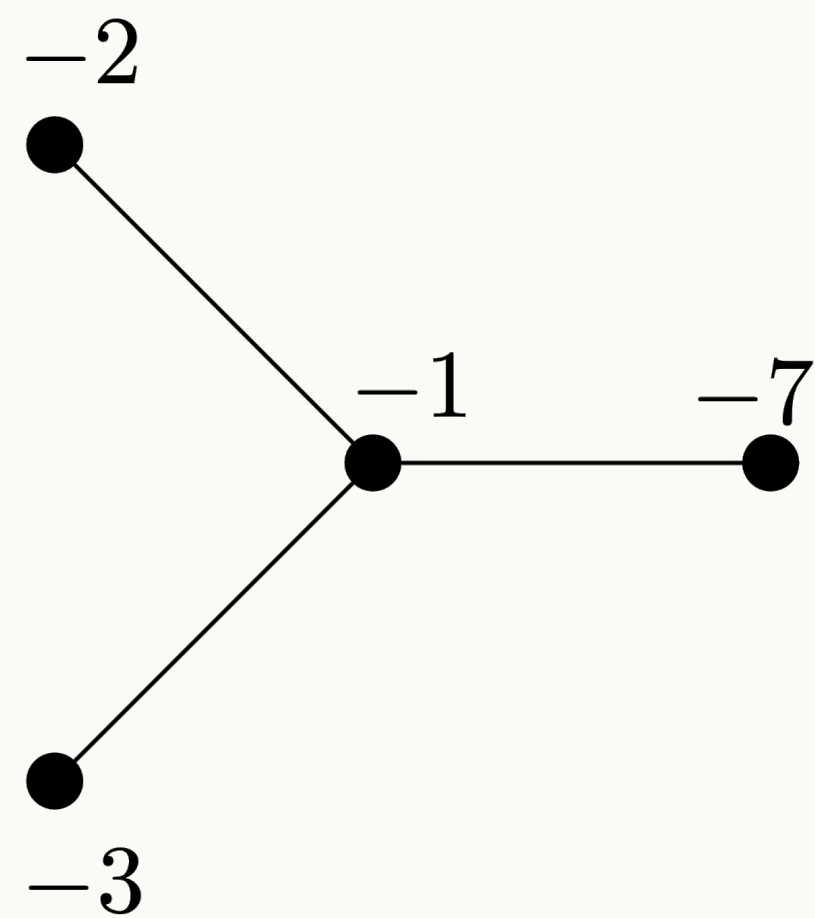
$$b \in (2\mathbb{Z}^s + \delta) / (2Q\mathbb{Z}^s)$$

$$b \in \text{Spin}^c(M)$$

Q : adjacency matrix, $\delta_v = \deg v$

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$$b \in (2\mathbb{Z}^s + \delta) / (2Q\mathbb{Z}^s)$$

$$\hat{Z}_b(M_3; q) = q^{\Delta_b} \int_{|z_v|=1} \prod_{v \in V} \frac{dz_v}{2\pi i z_v} \left(z_v - \frac{1}{z_v} \right)^{2-\deg v} \Theta_b(z, q)$$

$$\text{where } \Theta_b(z, q) = \sum_{n \in b + 2Q\mathbb{Z}^s} q^{-\frac{n^T Q^{-1} n}{4}} \prod_{v \in V} z_v^{n_v}$$

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Theorem (GPPV, GM): Let M be a plumbed manifold, represented by (Γ_M, Q) . Assume Q is weakly negative definite. Then, $\hat{Z}_b(\Gamma_M; q) \in q^{\Delta_b} \mathbb{Z}[[q]]$ does not depend on the representation of M as a plumbed manifold.

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Conjecture (JG): Let $\hat{Z}_b(q) = q^{\Delta_b} \sum_{n \geq 0} a_n q^n$. Then, $a_n \sim \text{Re} \left[\exp \left(2\pi \sqrt{\frac{1}{6} c_{eff} n} \right) \right]$, where $c_{eff} \in \mathbb{C}$.

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Theorem (CCFFGHP): There exist infinite families of plumbed manifolds M for which $\hat{Z}_b(M)$ corresponds to characters of the same logarithmic Vertex Operator Algebra for every $b \in \text{Spin}^c(M)$.

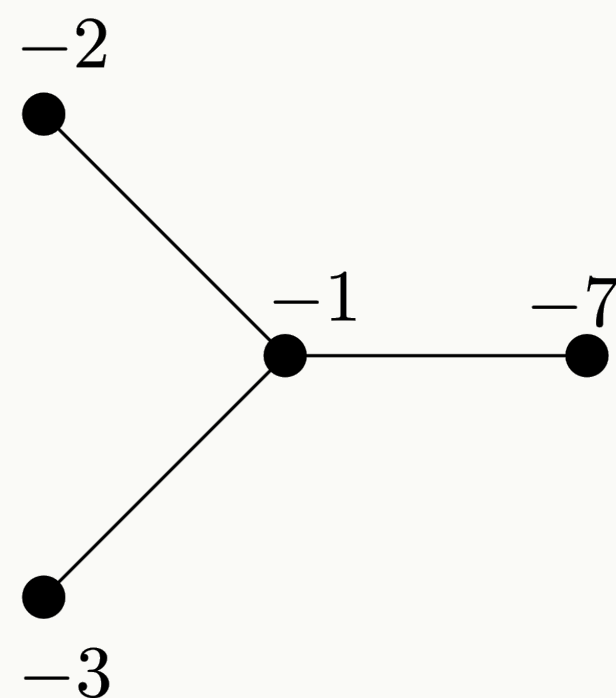
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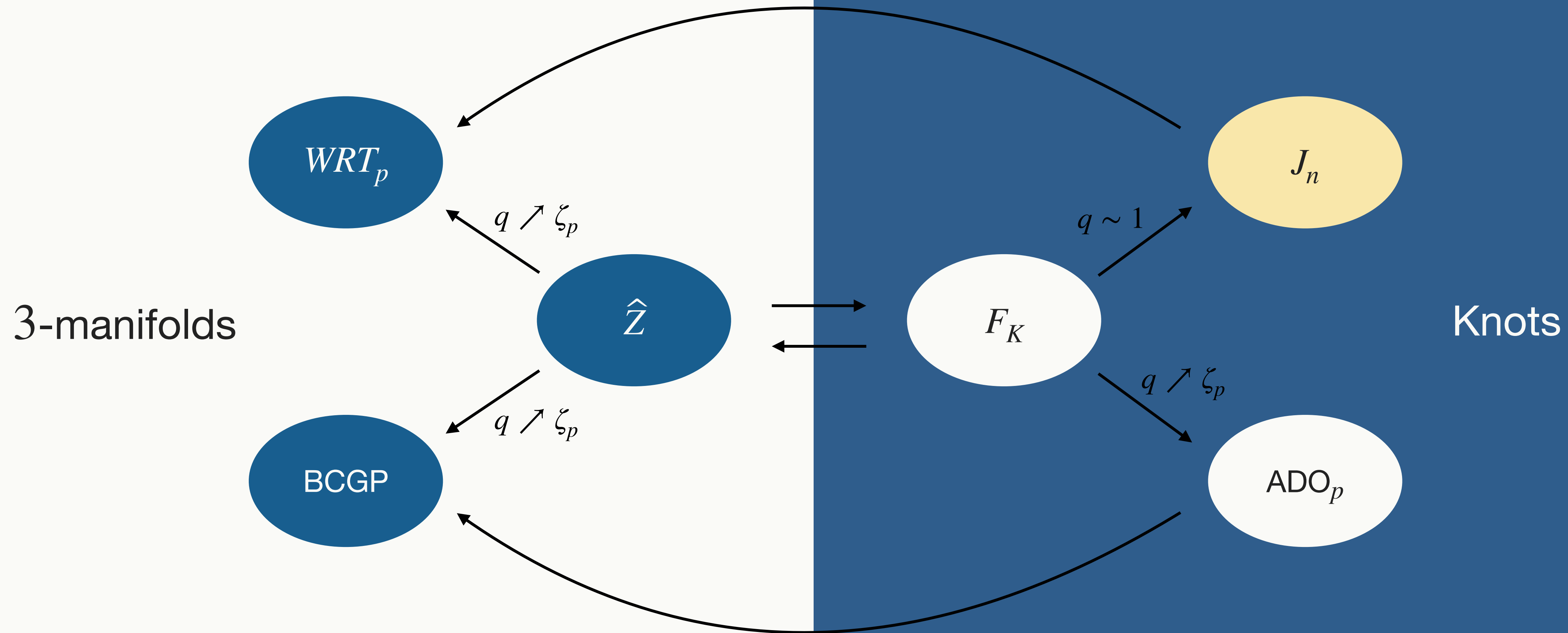
Theorem (CCFFGHP): There exist infinite families of plumbed manifolds M for which $\hat{Z}_b(M)$ corresponds to characters of the same logarithmic Vertex Operator Algebra (log-VOA) for every $b \in \text{Spin}^c(M)$.

Example: Brieskorn sphere $\Sigma(2,3,7)$.

$$\psi_{p,a}(\tau) := \sum_{k \equiv a \pmod{2p}} \text{sgn}(k) q^{\frac{k^2}{4p}}, \quad q = e^{2\pi i \tau}$$



$$\begin{aligned} \hat{Z}_0(q) &= q^{\Delta} \left(\psi_{42,1}(\tau) - \psi_{42,13}(\tau) - \psi_{42,29}(\tau) + \psi_{42,41}(\tau) \right) \\ &= \chi \left(M_{1,1} \oplus M_{1,13} \oplus M_{1,29} \oplus M_{1,41} \right) \\ &= -q^{-\frac{1}{2}} (1 + q + q^3 + q^4 + q^5 + 2q^7 + q^8 + \dots) \end{aligned}$$



Colored Jones invariants and VOAs

Quantum groups: $U_q(\mathfrak{sl}_2)\text{-rep}^{f.d.}$ is a **ribbon category**:

- Braiding between representations.
- Dual objects: pivotal map, evaluation and coevaluation maps.

$$\left(\beta = \text{diagram of two strands with four crossings}, V_n \right) \xrightarrow{\text{tr}^q} J_n(K; q) \in \mathbb{Z}[q, q^{-1}]$$

Denote $\delta_n := \min \deg J_n(q)$. Then, δ_n is a quadratic quasipolynomial in n .

Let $\Phi := \lim_{n \rightarrow \infty} q^{-\delta_n} J_n(q)$ whenever the limit is convergent in $\mathbb{Z}((q))$. We call Φ the **Jones tail**.

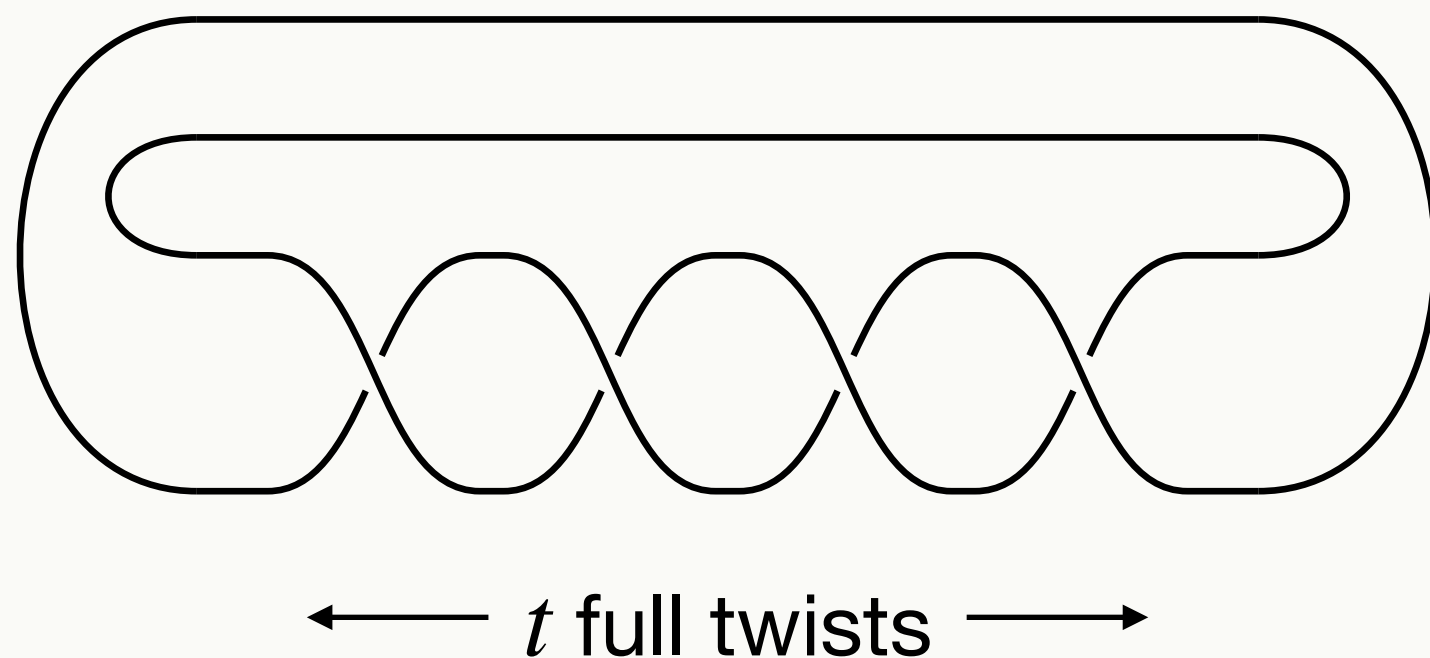
What do we know about the tail of colored Jones Φ ?

- Multiple tails, depending on the period of δ_n .
- Unique for alternating knots (Garoufalidis—Lê). Can be expressed in terms of Nahm sums.
- It does not exist for all knots (Lee—van der Veen).

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Example (**Bringmann—Milas**): $T(2,2t)$



1 full twist: $\Psi(q) = 1$

2 full twists: $\Psi(q) = 1 - q + q^3 - q^6 + q^{10} - q^{15} + q^{21} + \dots$

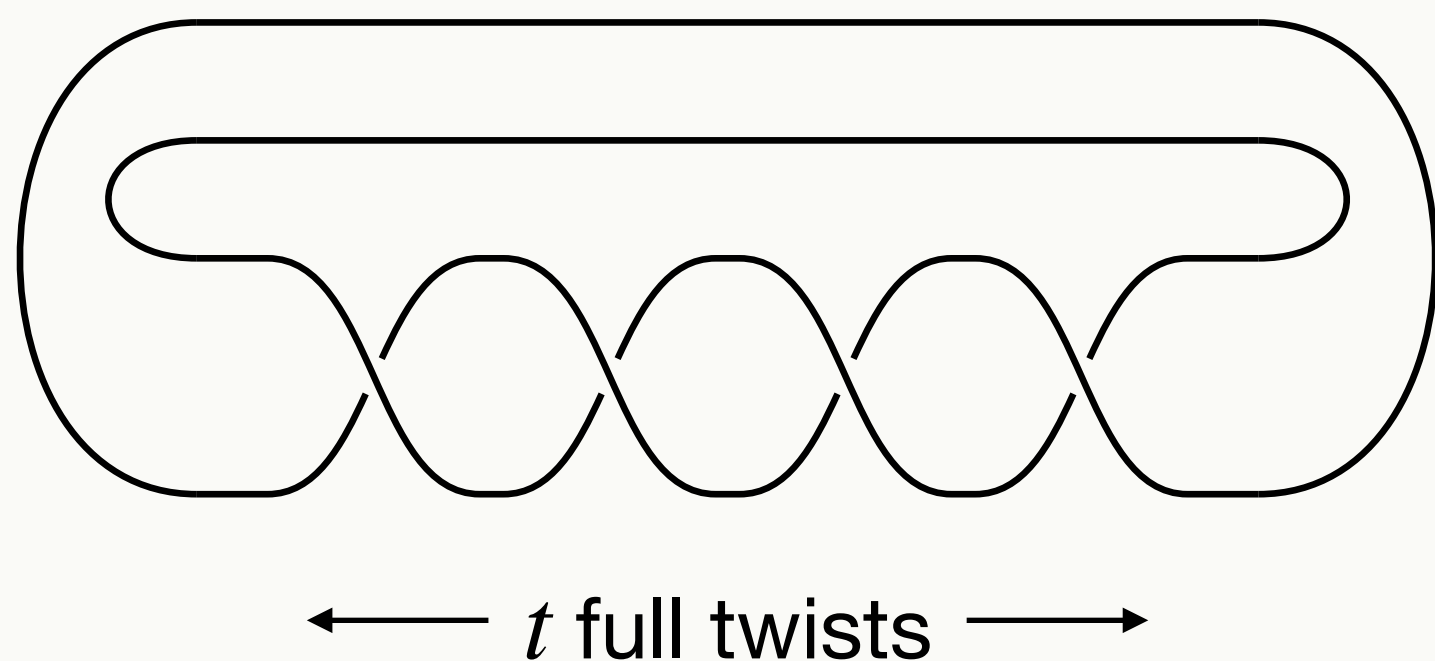
3 full twists: $\Psi(q) = 1 - q + q^5 - q^8 + q^{16} - q^{21} + q^{33} - q^{40} + \dots$

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Let $\psi_t(q) := \sum_{n \in \mathbb{Z}} \text{sgn}(n) q^{tn^2 + (t-1)n} \Rightarrow \psi_t(q)$ corresponds to a character of the **singlet $(1, t)$ log-VOA**.

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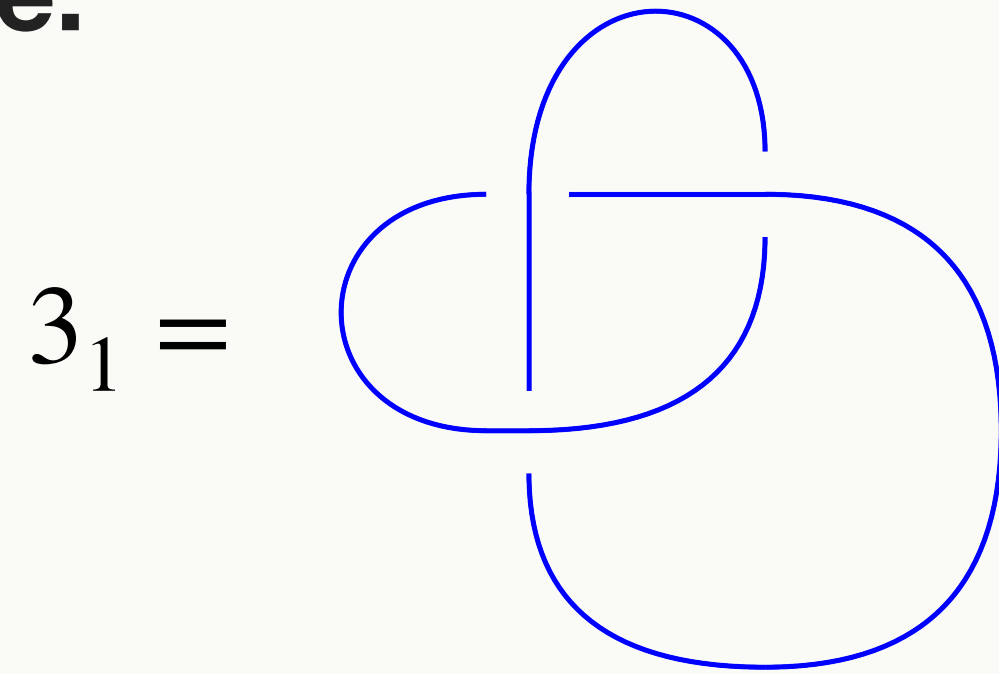
Quantum modular properties

- The tail of the torus links $T(2,2t)$ agrees with $\psi_t(q)$ (**Bringmann—Milas**).
- The tail of $T(2s,2t)$ recovers the character of the singlet (s,t) -log VOA (**Hikami—Sugimoto**).
- More generally, the tail of certain arborescent links are products of $\psi_t(q)$ and $\theta_b(q)$, where $\theta_b(q) := \sum_{n \in \mathbb{Z}} (-1)^n q^{\frac{bn(n+1)}{2} - n}$, b odd (**Osburn—Storzer**).

Perturbative behaviour of colored Jones

The set of all colored Jones polynomials $\{J_n(q)\}_{n \geq 0}$ presents a lot of structure.

Example:

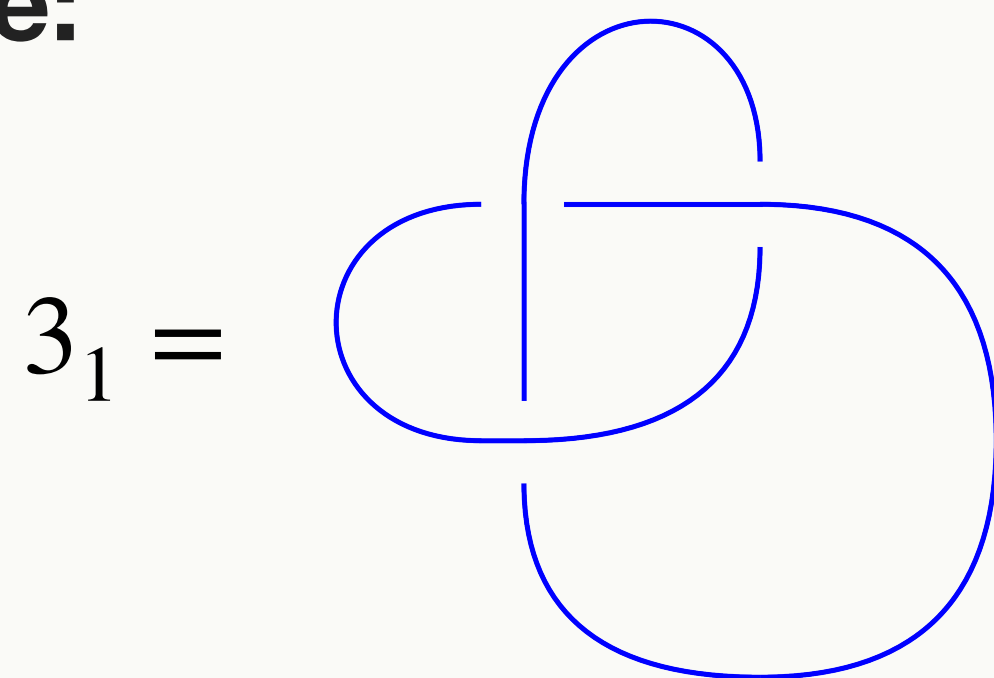


$$J_n(3_1; q) = \sum_{m \geq 0} (-1)^m q^{\frac{m(m+3)}{2}} \prod_{i=1}^m (q^n + q^{-n} - q^i - q^{-i})$$

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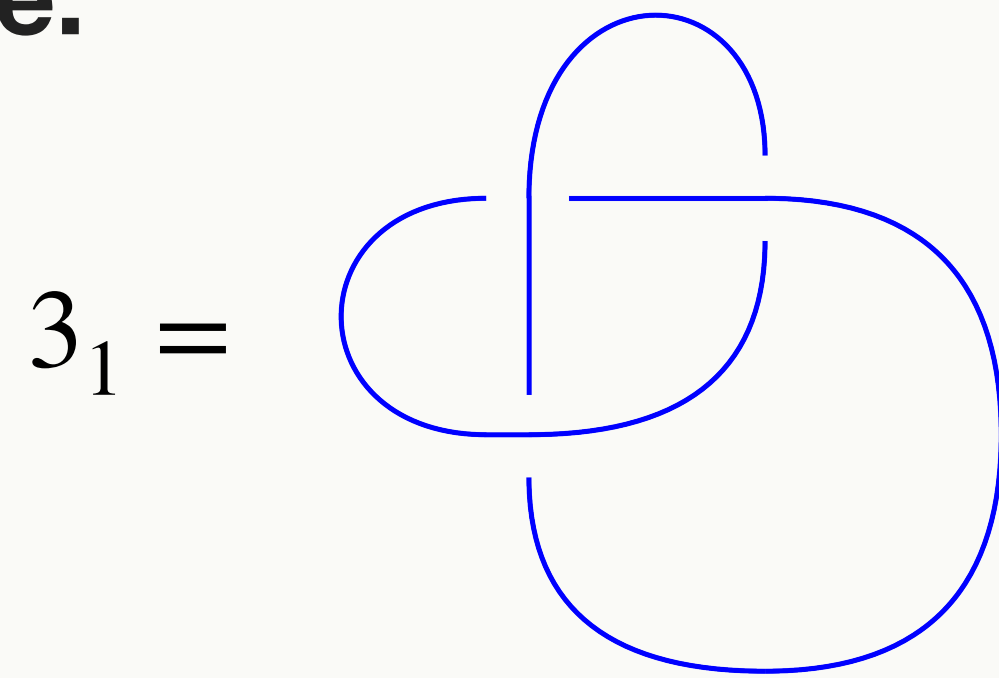
$$J_n(3_1; e^{\hbar}) = 1 + \frac{\hbar^2}{2!}(2 - 2n^2) + \frac{\hbar^3}{3!}(12 - 12n^2) + \frac{\hbar^4}{4!}(22n^4 - 168n^2 + 146) + \dots$$

at $\hbar \rightarrow 0$.

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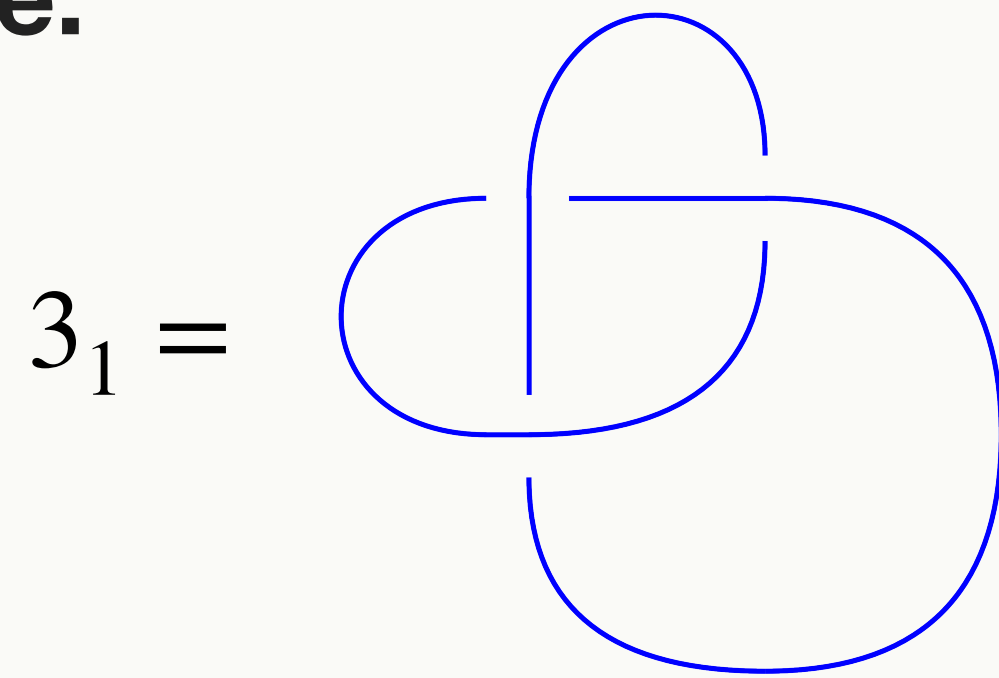
$$= \frac{x}{1 - x + x^2} + \frac{x(1 - 2x + 2x^2 - 2x^3 + x^4)}{(1 - x + x^2)^3} \hbar + \dots \Big|_{x=e^{\hbar n}} = \sum_{j \geq 0} \hbar^j \frac{P_j(x)}{\Delta^{2j+1}(x)} \Big|_{x=e^{\hbar n}}$$

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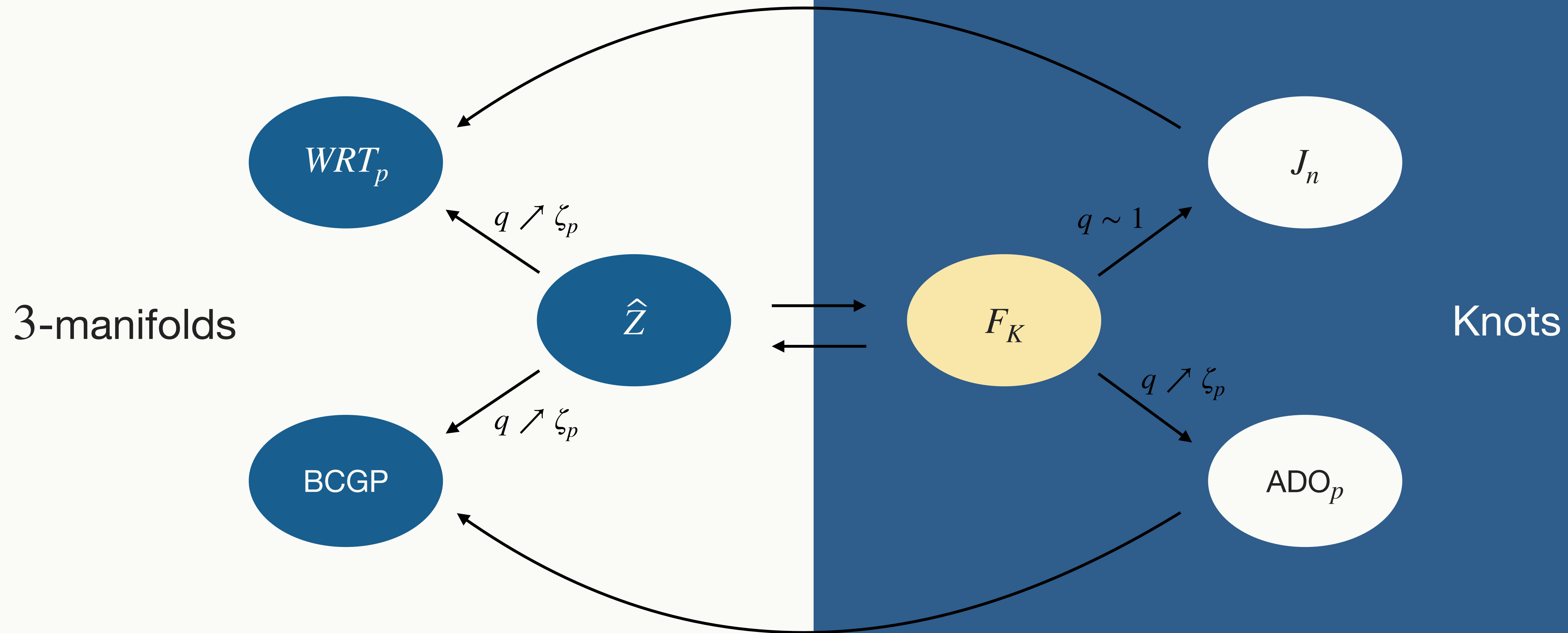
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at $\hbar \rightarrow 0$. Define $\Psi_K(x, \hbar) = \sum_{j \geq 0} \hbar^j \frac{P_j(x)}{\Delta^{2j+1}(x)} \in \mathbb{Z}(x)[[\hbar]]$.



F_K invariant

Gukov—Manolescu predict the existence of a two-variable series for every knot K

$$F_K(x, q) \in \mathbb{Z}((q))[[x]]$$

with the property that

$$F_K(x, e^{\hbar}) = \Psi_K(x, \hbar)$$

at x and $\hbar$ near 0.

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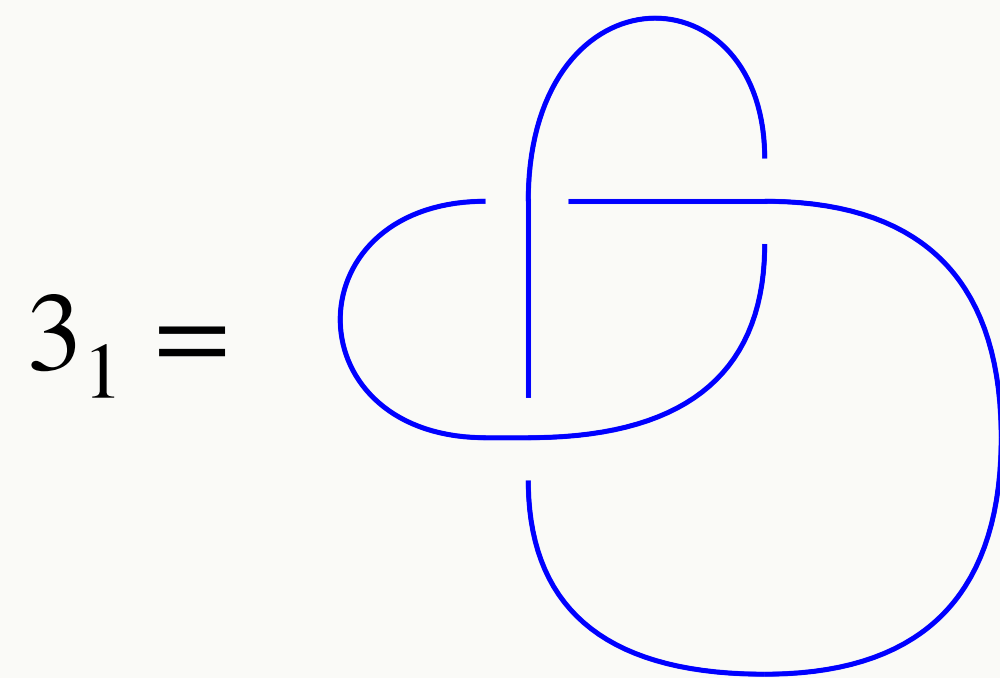
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Example: $F_{3_1}(x, q) = qx + qx^2 + (q - q^2)x^3 + (q - q^2 - q^3)x^4 + (q - q^2 - q^3)x^5 + \dots$



Then, for $\hbar \rightarrow 0$:

$$\begin{aligned} F_{3_1}(x, e^{\hbar}) &= (x + x^2 - x^4 - x^5 + x^7 + \dots) + (x + x^2 - x^3 - 4x^4 - 4x^5 + \dots)\hbar + \dots \\ &= \frac{x}{1 - x + x^2} + \frac{x(1 - 2x + 2x^2 - 2x^3 + x^4)}{(1 - x + x^2)^3}\hbar + \dots \end{aligned}$$

F_K invariant

Park's construction: Let V_λ^h (V_λ^l) be the **infinite-dimensional** highest (lowest) weight **Verma modules**, of highest (lowest) weight $\lambda \in \mathbb{C}$.

$$\left(\beta = \begin{array}{c} \text{diagram of two strands with four crossings} \end{array}, l, R^{inv}, V_{xq^{-1}}^h, V_{x^{-1}q}^l \right) \longmapsto F_K(x, q) \in \mathbb{Z}((q))[[x]]$$

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In the general case, F_K is not computed as a trace over $U_q(\mathfrak{sl}_2)$ modules.
 Instead, we use a **modified trace** called *inverted state sum* (hidden in ι, R^{inv}).
 We have yet to **find the true quantum algebra** sitting behind our computations.

F_K invariant

We have some hints:

Cyclotomic expansion of Jones
Habiro

$$J_n(q) = \sum_{m \geq 0} a_m(K; q) \sigma_m(q^n) = \sum_{m \geq 0} a_m(q) \prod_{j=1}^m (q^n + q^{-n} - q^j - q^{-j})$$

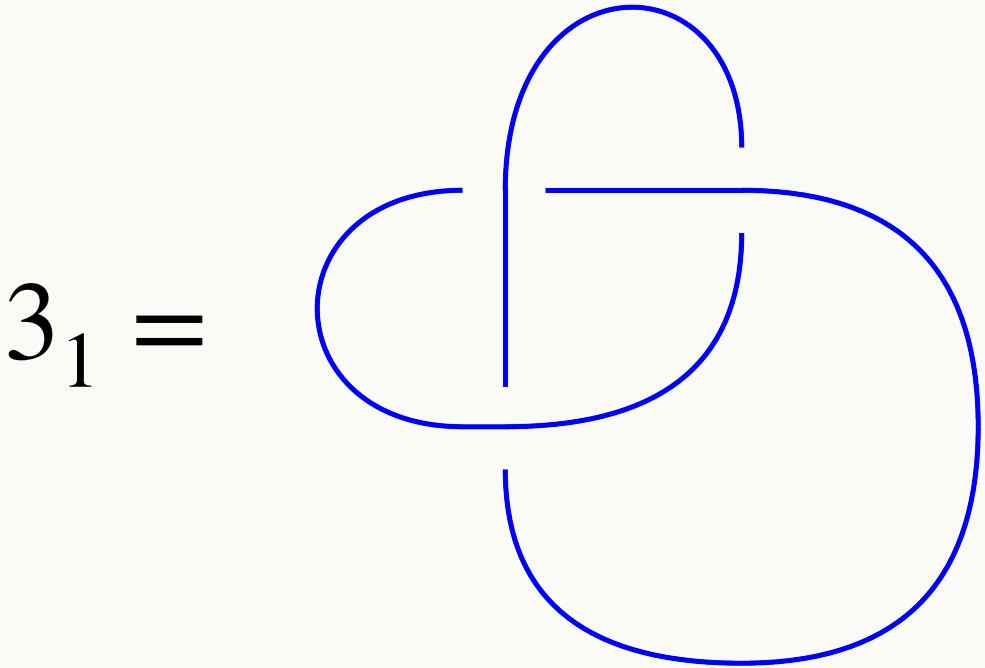
Inverted Habiro series
Park

$$F_K(x, q) = - \sum_{m > 0} a_{-m}(K; q) \sigma_{-m}(x) = - \sum_{m > 0} a_{-m}(q) \frac{1}{\prod_{j=0}^{m-1} (x + x^{-1} - q^j - q^{-j})}$$

The coefficients $\{a_n(q)\}_{n \geq 0}$ and $\{a_{-m}(q)\}_{m > 0}$ share similar structural properties.

Examples: $a_m(3_1) = (-1)^m q^{\frac{m(m+3)}{2}}$ and $a_m(4_1) = 1$ for all $n \in \mathbb{Z}$.

F_K invariant



Example: It was observed by Svoboda that, for $a_m(3_1) = (-1)^m q^{\frac{m(m+3)}{2}}$, both expansions are related via the celebrated Ramanujan identity:

$$\sum_{k \geq 0} \frac{q^k}{(1-x) \prod_{i=1}^k (1-xq^i)(1-x^{-1}q^i)} = \sum_{k=0}^{\infty} (-1)^k x^{3k} q^{k(3k+1)/2} (1-x^2 q^{2k+1}) + \frac{\sum_{k=0}^{\infty} (-1)^k x^{2k+1} q^{k(k+1)/2}}{(1-x) \prod_{i=1}^{\infty} (1-xq^i)(1-x^{-1}q^i)}$$

Inverted Habiro series for F_K

Power series expansion of F_K

Information of the residues at $x = q^n$

Study of the F_K invariant

In work with [Orland](#), [Passaro](#), [Saunders-A'Court](#) and [Svoboda](#), we compute and study F_K for two classes: fibered knots and strongly quasipositive knots.

F_K of fibered knots

Definition: A knot K is fibered if there exists a fibration of the knot complement $S^3 \setminus K$ over S^1 .

Theorem (OSSS): Let K be a homogeneous braid knot (and therefore fibered). Then,

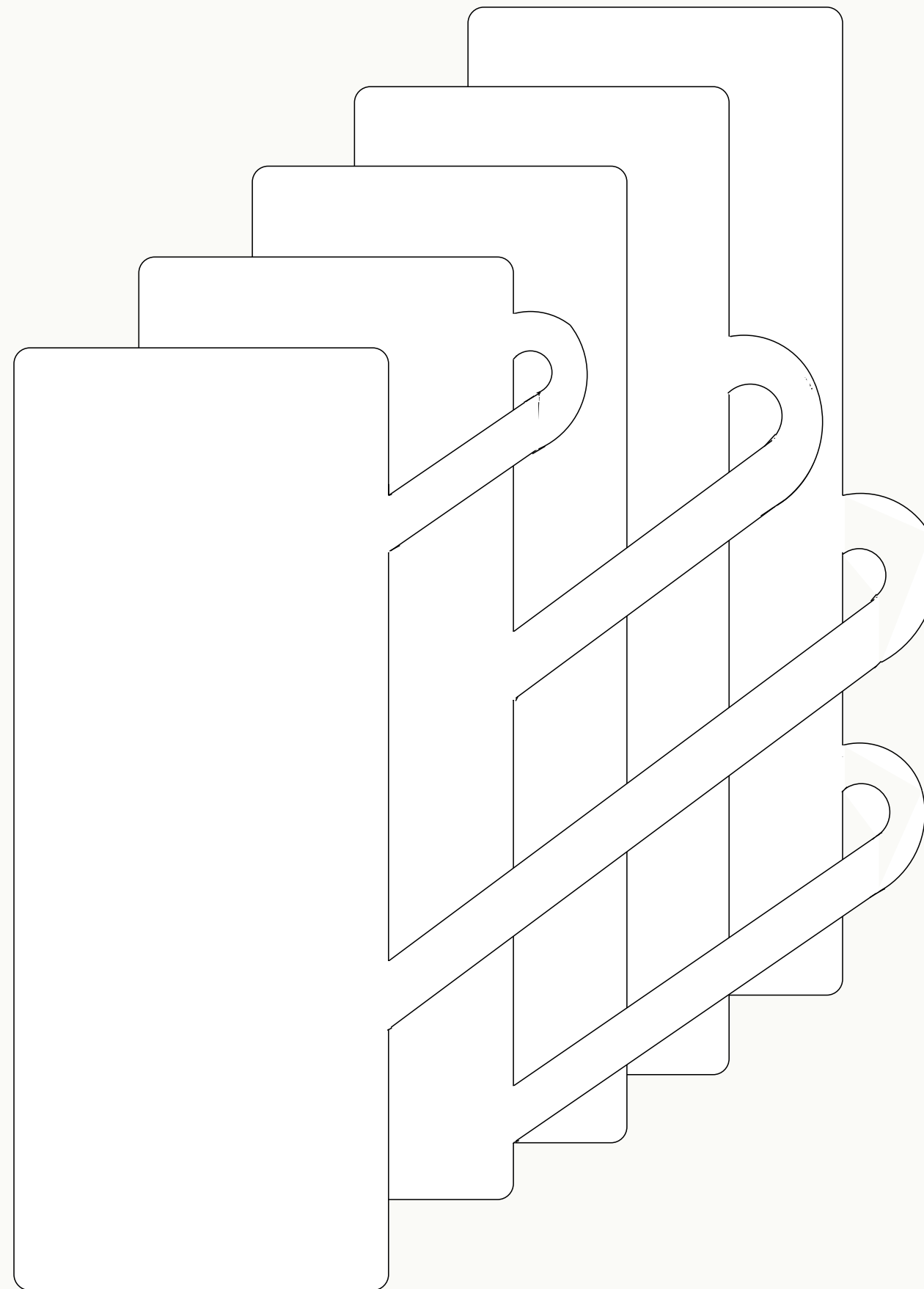
$$F_K(x, q) = \pm q^{\ell_K} x^{g_K} + (\text{higher order terms in } x) \in \mathbb{Z}[q, q^{-1}][[x]].$$

Theorem (OSSS): The same holds for the first 1200 fibered knots.

Study of the F_K invariant

F_K of strongly quasipositive (SQP) knots

Strongly quasipositive knots arise as the boundary of special surfaces.



Study of the F_K invariant

F_K of strongly quasipositive (SQP) knots

Strongly quasipositive knots arise as the boundary of special surfaces.

In this case, we are able to compute F_K using the highest weight Verma modules V_λ^h of $U_q(\mathfrak{sl}_2)$, whenever this is convergent in $\mathbb{Z}((q))[[x]]$.

$$\left(\beta = \text{[link diagram]}, V_{xq^{-1}}^h \right) \xrightarrow{\text{tr}^q} F_K(x, q) \in \mathbb{Z}((q))[[x]]$$

Study of the F_K invariant

Conjecture (OSSS): For a SQP knot K , the colored Jones $\{J_n(q)\}_{n \geq 0}$ determine $F_K(x, q)$.
Moreover, the Jones tail $\Phi(q)$ exists and agrees with $f_0(q)$, the lowest x -graded term in F_K :

$$f_0(q) = \Phi(q)$$

We observed this for the families $\{T(2, 2n)\}_{n \geq 0}$ and the first SQP knots.

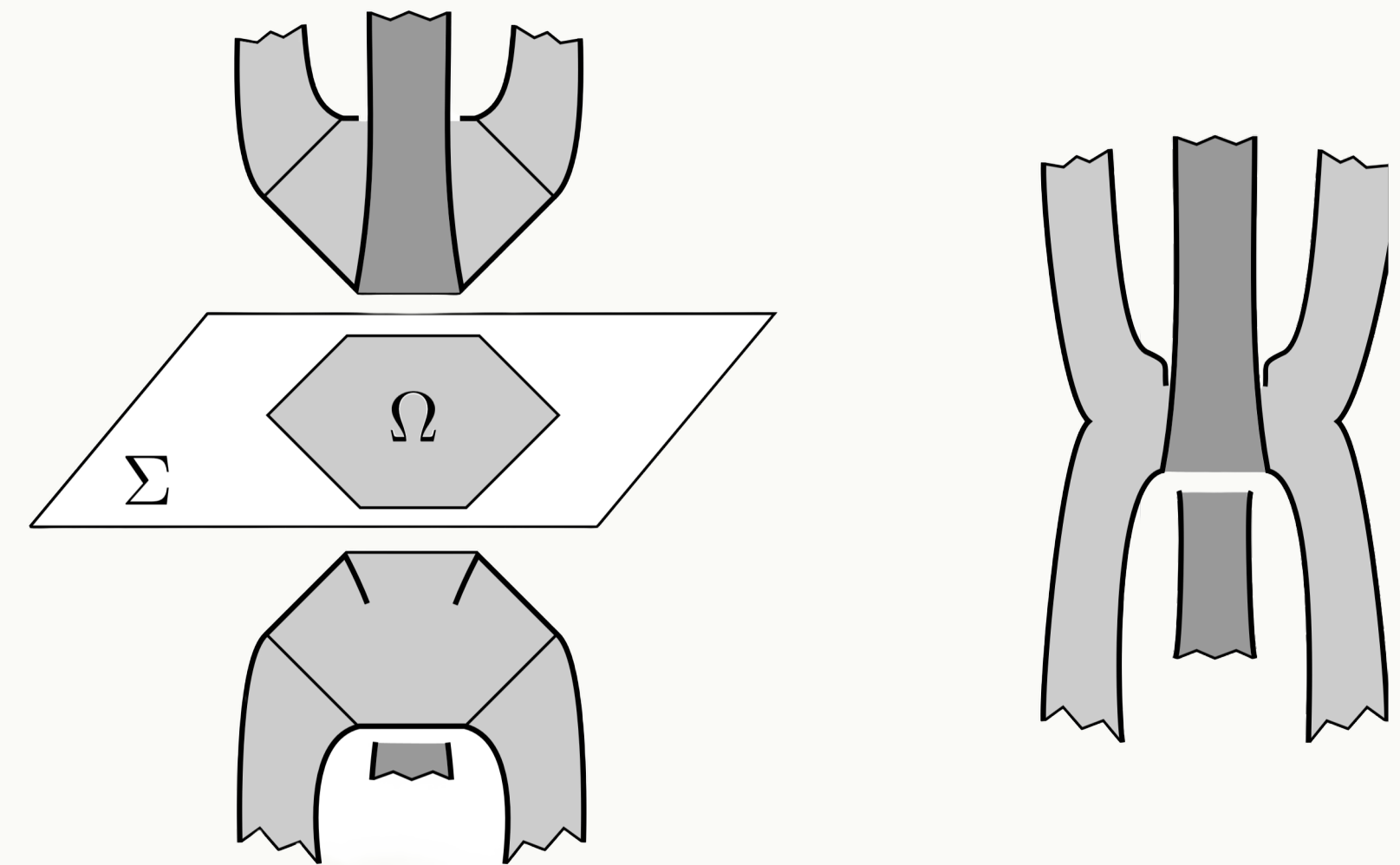
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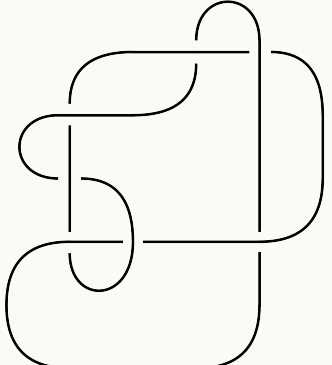
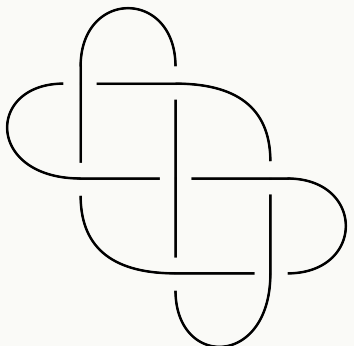
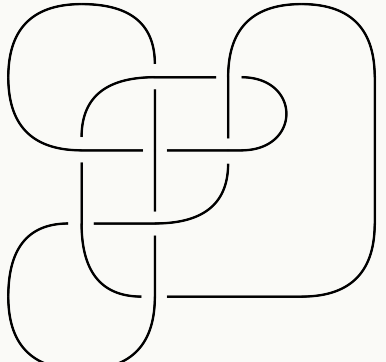
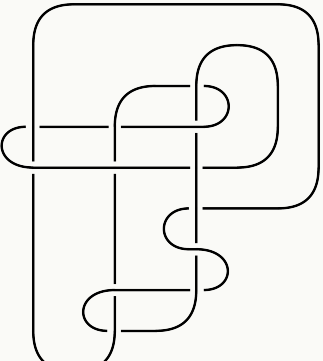
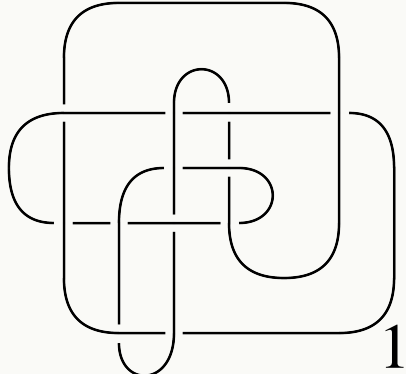
$$f_0(q) = \Phi(q)$$

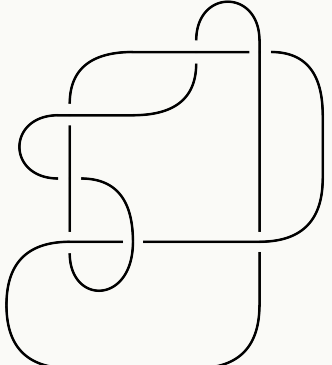
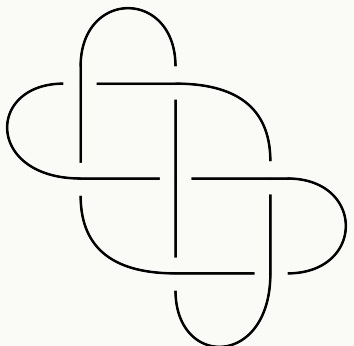
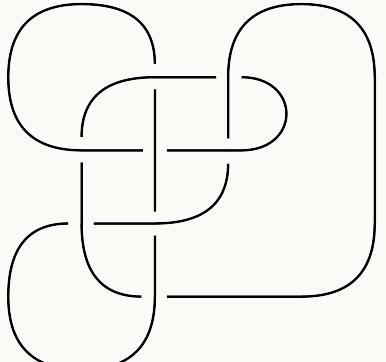
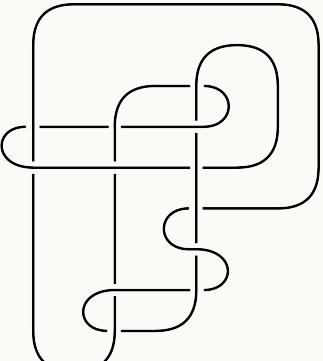
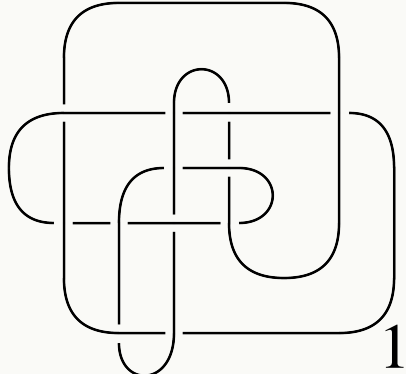
We observed this for the families $\{T(2, 2n)\}_{n \geq 0}$ and the first SQP knots.

Observation (OSSS): If a link L_3 can be obtained by plumbing two links L_1 and L_2 along minimal genus surfaces, then $f_0(L_3) = f_0(L_1) \cdot f_0(L_2)$.



$\Rightarrow$ We observe that $f_0(q)$ agrees with the characters of the **singlet $(1, p)$ log-VOA** for plumbings of $\{T(2, 2n)\}_{n \geq 0}$.

Knot	$f_0(q)$	Closed formulas
 7_2	$1 - q + q^5 - q^8 + q^{16} - q^{21} + \dots$	$\psi_3(q)$
 7_4	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\psi_2^2(q)$
 8_{15}	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\sum_{n \geq 0} (-1)^n q^{\frac{n(n+1)}{2}} \sum_{m=0}^n q^{-m(n-m)} \frac{(q^n, q^{-1})_m}{(q)_m}$
 $12n_{293}$	$1 + q + q^4 + q^9 + q^{16} + q^{25} + \dots$	$\sum_{n \geq 0} q^{n^2}$
 $12n_{801}$	$1 + 2q + q^2 + q^3 + 2q^4 + 2q^5 + 2q^6 + \dots$	

Knot	$f_0(q)$	Closed formulas
 7_2	$1 - q + q^5 - q^8 + q^{16} - q^{21} + \dots$	$\sum_{n \geq 0} (-1)^n q^{\frac{n(n+1)}{2}} \sum_{m=0}^n q^{-m(n-m)} \frac{(q^{-n}, q)_m}{(q^{-1})_m}$
 7_4	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\psi_2^2(q)$
 8_{15}	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\sum_{n \geq 0} (-1)^n q^{\frac{n(n+1)}{2}} \sum_{m=0}^n q^{-m(n-m)} \frac{(q^n, q^{-1})_m}{(q)_m}$
 $12n_{293}$	$1 + q + q^4 + q^9 + q^{16} + q^{25} + \dots$	$\sum_{n \geq 0} q^{n^2}$
 $12n_{801}$	$1 + 2q + q^2 + q^3 + 2q^4 + 2q^5 + 2q^6 + \dots$	

Limits of the F_K invariant as $q \rightarrow 1$

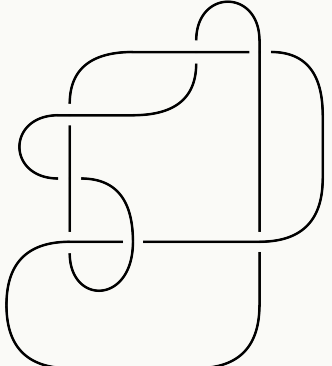
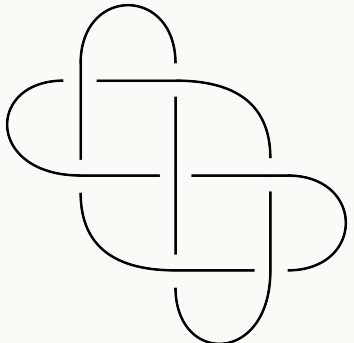
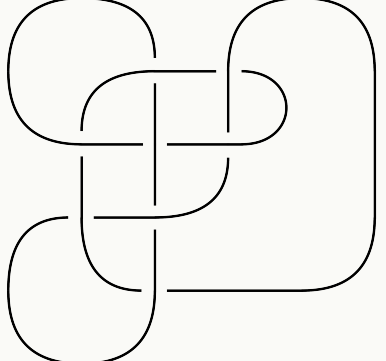
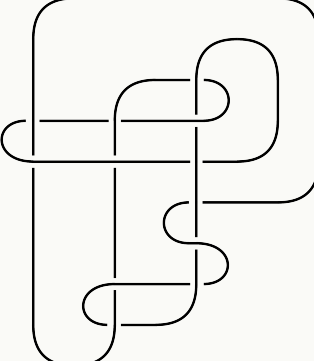
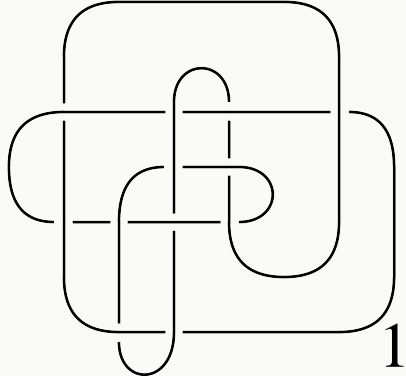
We observe three different behaviours:

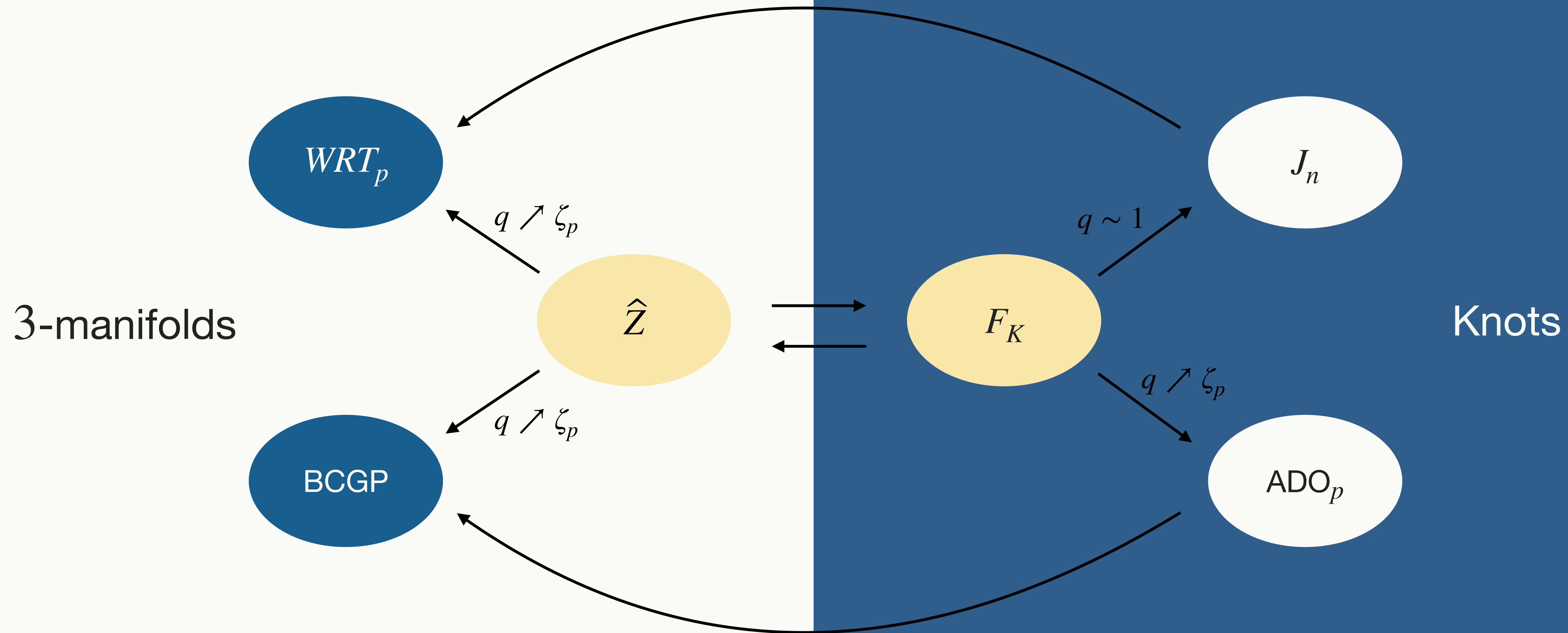
1. K is fibered and $\lim_{q \rightarrow 1} |f_0(q)| = |f_0(1)| = 1$.

2. K is a non-fibered, SQP knot and $\lim_{q \rightarrow 1} |f_0(q)| = \frac{1}{n}$ for some $n \in \mathbb{Z}_{>0}$.

3. K is a non-fibered, SQP knot and numerical fitting suggests that

$$\lim_{q \rightarrow 1} |f_0(q)| \sim \frac{\sqrt{\pi}}{\sqrt{-\log q}} \exp\left(\frac{a}{-\log q}\right), \text{ where we expect } a \text{ to be related to the topology of } K.$$

Knot	$f_0(q)$	Behaviour as $q \rightarrow 1$
 7_2	$1 - q + q^5 - q^8 + q^{16} - q^{21} + \dots$	$\frac{1}{3} + O(\log q)$
 7_4	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\frac{1}{4} + O(\log q)$
 8_{15}	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$\frac{1}{3} + O(\log q)$
 $12n_{293}$	$1 + q + q^4 + q^9 + q^{16} + q^{25} + \dots$	$\frac{\sqrt{\pi}}{\sqrt{-\log q}} + O(1)$
 $12n_{801}$	$1 + 2q + q^2 + q^3 + 2q^4 + 2q^5 + 2q^6 + \dots$	$\sim \frac{\sqrt{\pi}}{\sqrt{-\log q}} e^{-\frac{a}{\log q}} \quad a \sim 0.25$



Thank you!